

Literature Review

Graphs for Forecasting Google Trends Search Interest in Florida Agricultural Products

Prepared as a modeling-focused research note for defining nodes, edges, graph structures, and forecasting experiments.

Main research goal

Study whether Google Trends search interest for agricultural products in Florida can be modeled as interconnected time-series signals using graph neural networks (GNNs).

How to read this document

- The most important column in the literature matrix is **Node and edge definition**. This is where each paper becomes useful for your own graph design.
- For your project, the cleanest first graph is a **crop-product graph**: each node is one crop's weekly Google Trends time series in Florida; edges represent correlation, lagged influence, Granger causality, seasonality similarity, or learned adjacency.
- The next stronger design is a **crop-by-region graph**: each node is a crop-region pair, such as strawberry-Florida, strawberry-Tampa, blueberry-Florida, etc.
- A later, more publishable extension is a **heterogeneous graph** with crop, region, season, weather, event, and price nodes.

Search outcome

I did not find a paper that exactly combines Florida agriculture + Google Trends + GNN forecasting. That is a useful gap: existing work usually covers (1) graph forecasting for traffic/weather/finance/sensors, (2) GNNs for agricultural yield or soil moisture, or (3) Google Trends/econometric forecasting without explicit graph neural modeling.

1. Research framing

The proposed study can be framed as a graph-based multivariate time-series forecasting problem. Each agricultural search term is treated not only as an independent time series, but as part of a system where products may move together because of seasonality, consumer substitution, shared growing seasons, weather, holidays, news, recipes, farm events, or price/demand attention.

Core modeling question

Can modeling crop search interest as a graph improve forecasting compared with non-graph baselines such as seasonal naive, ARIMA/SARIMA, VAR, LSTM, Temporal CNN, or Transformer models?

Possible Florida crop nodes

Suggested crops/products: strawberry, blueberry, tomato, orange, watermelon, sweet corn, cucumber, bell pepper, snap beans, cabbage, lettuce, eggplant, peach, avocado, citrus, pumpkin, peanut, nursery plants, and ornamental plants. For a first experiment, keep the graph small and interpretable: 8-15 nodes is enough.

Why graph modeling is appropriate

- Google Trends search interest is multivariate: one crop's demand-attention signal may lead, lag, substitute for, or co-move with another crop's signal.
- Agricultural products have shared seasonal calendars, weather sensitivity, and consumer behavior patterns, which can be encoded as edges.
- GNN forecasting models let the model borrow information across related time series instead of forecasting each crop separately.
- Learned-adjacency methods such as MTGNN, AGCRN, Graph WaveNet, StemGNN, TimeGNN, and SDGL are useful when the true graph is unknown.

Recommended novelty statement

Existing graph forecasting studies mainly focus on traffic, weather, finance, energy, and sensor-based agriculture. Existing Google Trends forecasting studies usually treat search variables as independent predictors or standard multivariate covariates. This study models agricultural Google Trends search interest as an interconnected graph of crop-level demand-attention signals, comparing statistical, econometric, semantic, seasonal, and learned graph structures for forecasting Florida agricultural product search interest.

2. Literature review matrix: how nodes and edges are defined

This table focuses on the modeling aspect: what becomes a node, how edges are constructed, and what lesson transfers to the Florida Google Trends agriculture project.

Paper / area	Node definition	Edge definition	Modeling idea	Transfer to Florida Google Trends
DCRNN - traffic forecasting [1]	Traffic sensors or road locations.	Directed road-network edges; diffusion process modeled with bidirectional random walks.	Diffusion graph convolution plus encoder-decoder recurrent forecasting.	Useful if you define crop-region nodes and want directed influence or diffusion-like attention flow across related regions/products.
STGCN - traffic forecasting [2]	Traffic stations/sensors.	Predefined spatial graph based on road structure or distance.	Graph convolution captures spatial dependency; temporal convolution captures time dynamics.	Template for a fixed graph: crop nodes connected by domain-defined similarity such as same season/category.
Graph WaveNet [3]	Sensors or variables in a spatial-temporal system.	Predefined graph plus learned adaptive dependency matrix from node embeddings.	Adaptive graph convolution plus dilated temporal convolution for long-range sequences.	Very relevant: combine hand-built crop similarity graph with a learned crop-crop dependency matrix.
MTGNN [4]	Each variable/time series is a node.	Learns directed inter-variable relations automatically; external attributes can be added.	Graph learning module + mix-hop propagation + dilated temporal convolution.	Closest baseline for your first crop-product graph. Nodes = crop search series; edges = learned crop dependencies.
AGCRN [5]	Traffic series or locations.	No predefined graph required; data-adaptive graph generation infers dependencies.	Node-adaptive parameter learning plus adaptive graph convolutional recurrent network.	Good if you want to avoid hand-designed crop edges and let the model infer relationships from Trends data.
StemGNN [6]	Each multivariate time-series component is a node.	Inter-series correlations learned in spectral domain, not from predefined priors.	Graph Fourier Transform for inter-series correlation plus Discrete Fourier Transform for temporal dependency.	Useful if crop search signals have strong seasonal/frequency patterns.
MTHetGNN [7]	Each variable is a node.	Multiple edge types: static, dynamic, predictable, and latent relations.	Heterogeneous graph embedding plus temporal CNN modules.	Strong inspiration for a multi-edge crop graph: correlation edge, season edge, price edge, weather edge, and learned edge.
SDGL [8]	Variables in multivariate time series.	Static graph for long-term patterns; dynamic graph for short-term time-varying relations.	Learns both static and dynamic graph matrices from data.	Important because crop relationships may change by season. Example: strawberry-blueberry relation may differ in winter vs spring.
TimeGNN [9]	Multiple time series as nodes.	Dynamic temporal graph representations capture evolving inter-series patterns.	Scalable dynamic graph learning for faster inference.	Good later-stage model if you have many crops, regions, or related search keywords.
Graph learning from dependent time series [10]	Each time-series component is a node.	Edges represent conditional dependence inferred from precision matrix/sparse-group lasso.	Statistical/econometric graph construction before forecasting.	Excellent non-neural graph-construction baseline: build edges from partial correlation or conditional dependence.
Macroeconomic forecasting with validated KGs [11]	News/event themes are nodes.	Edges are statistically significant co-occurrence/association relationships.	Graph backbone + centrality features improve macroeconomic forecasting.	Supports adding event/news/season nodes if you later include narratives around Florida agriculture, inflation, holidays, or weather shocks.
THGNN - financial time series [12]	Stocks/companies are nodes.	Daily dynamic relation graph generated from historical price movements.	Transformer encodes temporal patterns; heterogeneous graph attention models dynamic relations.	Useful analogy: crops behave like financial assets in a dynamic attention market; edges can update weekly.
Dynamic soil moisture GNN [13]	Soil moisture measurement locations or spatial units.	Temporal graph structures learned from data; handles missing ground truth.	End-to-end GNN learns changing spatial-temporal relations.	Agriculture-specific evidence that graph learning helps when physical relationships are hard to define exactly.

Paper / area	Node definition	Edge definition	Modeling idea	Transfer to Florida Google Trends
Domain-inspired temporal GCN for soil moisture [14]	Hydrological response units.	Graph topologies built through domain-inspired clustering by hydrological properties, not only spatial distance.	Temporal GCN plus GRU for soil moisture forecasting.	Very relevant principle: do not connect crops only by correlation; also connect by domain knowledge such as season, commodity group, and Florida production calendar.
Agri-GNN for yield prediction [15]	Farming plots are nodes.	Edges based on spatial and genotypic similarity.	GraphSAGE aggregates plot information through genotypic-topological filters.	Shows how agriculture graph edges can be built from domain similarity. For your case: product category, seasonality, and region similarity.
Graph sparsification for crop yield [16]	Field/county crop-yield units.	Starts from dense graph and removes edges while preserving graph properties via Fiedler-number-based sparsification.	Sparse GCN for faster crop yield prediction.	Helpful if crop graph becomes too dense. Compare dense correlation graph vs sparsified graph.
HiSTGNN for weather forecasting [17]	Meteorological variables and stations are nodes in a hierarchical graph.	Variable-level and station-level adaptive graphs; cross-regional interactions.	Hierarchical spatio-temporal GNN with adaptive graph learning and dilated temporal convolution.	Useful for crop-region design: product-level graph nested inside region-level graph.
Strawberry yield/price forecasting [18]	Not graph-based; uses station soil features, satellite images, and time series.	No explicit graph; relationships are learned through CNN-LSTM/GRU/attention ensembles.	Forecasts strawberry yields and prices using satellite and soil data.	Directly relevant crop example. Could become a baseline or covariate source for Florida strawberry search interest.
Supply-chain production-function GNN [19]	Firms are nodes.	Transactions between firms are edges; production functions are hidden relationships.	Temporal GNN plus inventory/production module forecasts future transactions.	Economics analogy: crops/markets form a demand-attention network; hidden relationships can be learned rather than fully specified.
Google Trends calibration [20]	Search queries are time series, not graph nodes in the original method.	No graph; calibration aligns multiple query time series onto a common scale using anchor terms.	Google Trends Anchor Bank addresses normalization/rounding issues.	Important preprocessing before graph learning, especially when comparing many crop terms.
Google Trends preprocessing [21]	Search queries are forecasting covariates.	No graph; preprocessing handles zeros, noise, sampling variability, and trend artifacts.	Hierarchical clustering, smoothing, and detrending improve forecasting signal.	Critical for your data pipeline: repeated pulls, averaging, smoothing, detrending, and missing/zero handling.
Aggregating Google Trends [22]	Many search queries become multivariate predictors.	No graph; combines/aggregates search volume data consistently across series.	PCA and aggregation improve signal-to-noise and reduce overfitting for economic nowcasting.	Useful baseline: compare graph model against PCA/VAR/ARIMAX with aggregated crop search terms.
Google Trends predictability in finance [23]	Search query series and price-return series.	No graph; tests predictive value with careful backtesting.	Warns about keyword selection bias and backtesting pitfalls.	Important for honest evaluation: define keywords before testing and avoid selecting crops only after seeing results.

3. Candidate graph designs for the Florida Google Trends project

Below are the most practical graph formulations. My recommendation is to begin with Design A because it is simple, interpretable, and directly aligned with multivariate time-series GNN papers like MTGNN, AGCRN, StemGNN, SDGL, and TimeGNN.

Design	Nodes	Edges	Node features at week t	Forecasting target	Best first models
A. Crop-product graph	One node per Florida crop/product search term: strawberry, blueberry, tomato, orange, watermelon, etc.	Correlation, lagged correlation, Granger causality, DTW similarity, shared season/category, or learned adjacency.	Google Trends score; lags 1, 2, 4, 8, 12 weeks; rolling mean; rolling std; month/season encoding; optional price/weather.	Next 1, 2, 4, or 8 weeks of Google Trends score for each crop.	MTGNN, AGCRN, Graph WaveNet, StemGNN, SDGL.
B. Crop-region graph	Crop-region pairs: strawberry-Florida, strawberry-Tampa, strawberry-Orlando, blueberry-Florida, etc.	Same crop across regions; different crops in same region; nearby regions; similar temporal patterns; seasonal edges.	Regional Google Trends score, local lag features, region/county embeddings, season features, optional weather.	Future search interest per crop-region node.	DCRNN/STGCN if region graph is fixed; Graph WaveNet/AGCRN if learned.
C. Heterogeneous graph	Multiple node types: Crop, Region, Season, Weather, Event, Price/Market.	Crop-crop, crop-region, crop-season, crop-weather, crop-event, region-region, price-crop relations.	Type-specific features: Trends for crops, weather variables, event indicators, season embeddings, price features.	Future crop search interest; optionally event-aware or price-aware forecasts.	MTHetGNN, heterogeneous GAT, temporal heterogeneous GNN.
D. Dynamic graph	Same as A or B, but graph changes by time window.	Edges recalculated over rolling windows or learned per time step.	Past window features plus time-varying adjacency matrix.	Short-horizon and seasonal-horizon forecasts.	SDGL, TimeGNN, THGNN-like dynamic graph modeling.

Recommended starting graph

Start with Design A: crop-product graph. It is easy to explain, easy to implement, and directly connects to multivariate time-series graph literature. Use weekly Google Trends values for Florida only. Treat each crop as a node and predict each crop's future search score.

4. Edge construction options

Edge type	How to compute it	Directed?	Why useful	Potential issue
Pearson/Spearman correlation	Compute correlation between crop search time series across the training period.	Usually no	Simple, interpretable, strong baseline.	May capture shared seasonality but not true influence.
Lagged correlation	Compute $\text{corr}(x_i(t-k), x_j(t))$ for several lags k .	Can be directed	Captures lead-lag relationships, e.g., one crop trend rising before another.	Needs careful lag selection and train-only calculation.
Granger causality	Test whether past values of crop i improve prediction of crop j beyond j 's own past.	Yes	Econometrically meaningful directed edges.	Can be unstable with short/noisy weekly data.
Dynamic Time Warping similarity	Measure shape similarity even if seasonal peaks are shifted.	Usually no	Good for crops with similar but shifted seasonal behavior.	Can create dense graphs; may need top-k pruning.
Season/category similarity	Connect crops that share UF/IFAS season windows, crop category, commodity group, or consumer use.	Usually no	Adds domain knowledge instead of relying only on noisy search signals.	Requires careful manual or external classification.
Learned adjacency	Let models such as MTGNN, AGCRN, Graph WaveNet, StemGNN, or SDGL learn edges end-to-end.	Depends on model	Useful when the true graph is unknown.	Less interpretable unless analyzed after training.
Hybrid graph	Start with domain/correlation edges and allow model to learn extra hidden edges.	Both possible	Best balance of interpretability and model flexibility.	More ablations needed.

5. Recommended experimental pipeline

- 1 Collect weekly Google Trends data for Florida for selected crops. Use a consistent date range, for example 2021-01-01 to 2025-12-31 or 2020-01-01 to 2025-12-31.
- 2 Use repeated pulls and averaging where possible, because Google Trends data can vary across downloads. Consider anchor terms/calibration if comparing many crops.
- 3 Create two data layouts: wide format where rows are weeks and columns are crops; long format with columns week, crop, google_trends_score, region, signal.
- 4 Build several graph variants using training data only: G1 correlation graph, G2 lagged-correlation graph, G3 Granger-causality graph, G4 season/category graph, and G5 learned adaptive graph.
- 5 Train non-graph baselines first: seasonal naive, ARIMA/SARIMA, VAR, LSTM, Temporal CNN, and Transformer/Informer if appropriate.
- 6 Train graph models: MTGNN and AGCRN first; then Graph WaveNet, StemGNN, SDGL, or TimeGNN if time permits.
- 7 Evaluate 1-week and 4-week ahead forecasts first. Then extend to 8-week horizon if the data are stable.
- 8 Report MAE, RMSE, MAPE/sMAPE, and per-crop error. Also inspect learned edges to see if they make agricultural sense.

Suggested first experiment

Use 10-12 Florida crop/product Google Trends series as nodes. Input window = past 12 weeks. Forecast horizon = next 1 week and next 4 weeks. Compare VAR, LSTM, and MTGNN/AGCRN. Use the same train/validation/test split across all models. For graph construction, compute edges only from the training period to avoid leakage.

Ablation plan

- No graph baseline: LSTM/TCN/Transformer on multivariate time series.
- Static correlation graph: Pearson or Spearman edges with top-k pruning.
- Econometric graph: Granger-causality edges.
- Domain graph: same season/category edges.
- Hybrid graph: domain + correlation edges.
- Learned graph: MTGNN/AGCRN/Graph WaveNet adaptive adjacency.
- Dynamic graph: rolling-window correlation or SDGL/TimeGNN-style dynamic relations.

Data leakage checklist

- Do not compute correlation, Granger, scalars, or PCA using the test period.
- Do not choose crop keywords based on test performance; define them before modeling.
- Use time-based split, not random split, because this is forecasting.
- Keep the same split and seed across graph and non-graph models.
- Report both aggregate accuracy and crop-level performance.

6. Google Trends-specific cautions

Google Trends is useful, but it is not a clean demand variable. It is a relative, sampled, normalized search-interest measure. The modeling pipeline should treat it as a noisy attention signal rather than direct sales or demand.

- Repeated downloads can differ because Google Trends uses samples. Pull multiple times and average when practical.
- Search scores are normalized to 0-100 within a request, so comparing many crops requires care.
- Low-volume queries can produce zeros or unstable signals. Prefer common search terms or Google Trends topics where available.
- Anchor-term calibration can help put many query series on a comparable scale.
- Smoothing and detrending can improve forecasting usefulness, but keep preprocessing fitted on the training period when evaluating forecasts.

Interpretable result to aim for

A strong paper/result would not only report better forecasting error. It would also show that learned or constructed crop relationships are meaningful. For example, strawberry may connect with blueberry through berry-season behavior; watermelon may connect with summer/holiday patterns; tomato and bell pepper may connect through vegetable/garden interest; citrus/orange may show different long-term behavior.

How to explain the project in one paragraph

This project treats Google Trends search interest for Florida agricultural products as a network of related time series. Instead of forecasting each crop independently, each crop is represented as a node in a graph, and edges represent relationships such as seasonality, correlation, lagged influence, product similarity, or learned dependence. Graph neural networks can then use information from neighboring crop signals to forecast future search interest. The study compares multiple graph definitions against standard time-series baselines to determine whether graph structure improves forecasting and whether the learned crop relationships are interpretable.

References and papers reviewed

The references below are included as working sources for the modeling review. Most links are arXiv records used to identify the modeling assumptions and graph definitions.

No.	Reference
[1]	Li, Y.; Yu, R.; Shahabi, C.; Liu, Y. Diffusion Convolutional Recurrent Neural Network: Data-Driven Traffic Forecasting. arXiv:1707.01926. https://arxiv.org/abs/1707.01926
[2]	Yu, B.; Yin, H.; Zhu, Z. Spatio-Temporal Graph Convolutional Networks: A Deep Learning Framework for Traffic Forecasting. arXiv:1709.04875. https://arxiv.org/abs/1709.04875
[3]	Wu, Z.; Pan, S.; Long, G.; Jiang, J.; Zhang, C. Graph WaveNet for Deep Spatial-Temporal Graph Modeling. arXiv:1906.00121. https://arxiv.org/abs/1906.00121
[4]	Wu, Z.; Pan, S.; Long, G.; Jiang, J.; Chang, X.; Zhang, C. Connecting the Dots: Multivariate Time Series Forecasting with Graph Neural Networks. arXiv:2005.11650. https://arxiv.org/abs/2005.11650
[5]	Bai, L.; Yao, L.; Li, C.; Wang, X.; Wang, C. Adaptive Graph Convolutional Recurrent Network for Traffic Forecasting. arXiv:2007.02842. https://arxiv.org/abs/2007.02842
[6]	Cao, D. et al. Spectral Temporal Graph Neural Network for Multivariate Time-series Forecasting. arXiv:2103.07719. https://arxiv.org/abs/2103.07719
[7]	Wang, Y. et al. MTHetGNN: A Heterogeneous Graph Embedding Framework for Multivariate Time Series Forecasting. arXiv:2008.08617. https://arxiv.org/abs/2008.08617
[8]	Li, Z.; Zhang, G.; Xu, L.; Yu, J. Dynamic Graph Learning-Neural Network for Multivariate Time Series Modeling. arXiv:2112.03273. https://arxiv.org/abs/2112.03273
[9]	Xu, N.; Kosma, C.; Vazirgiannis, M. TimeGNN: Temporal Dynamic Graph Learning for Time Series Forecasting. arXiv:2307.14680. https://arxiv.org/abs/2307.14680
[10]	Tugnait, J. K. Graph Learning from Multivariate Dependent Time Series via a Multi-Attribute Formulation. arXiv:2205.00007. https://arxiv.org/abs/2205.00007
[11]	Tilly, S.; Livan, G. Macroeconomic Forecasting with Statistically Validated Knowledge Graphs. arXiv:2104.10457. https://arxiv.org/abs/2104.10457
[12]	Xiang, S. et al. Temporal and Heterogeneous Graph Neural Network for Financial Time Series Prediction. arXiv:2305.08740. https://arxiv.org/abs/2305.08740
[13]	Vyas, A.; Bandyopadhyay, S. Dynamic Structure Learning through Graph Neural Network for Forecasting Soil Moisture in Precision Agriculture. arXiv:2012.03506. https://arxiv.org/abs/2012.03506
[14]	Azmat, M. et al. Forecasting Soil Moisture Using Domain Inspired Temporal Graph Convolution Neural Networks To Guide Sustainable Crop Management. arXiv:2212.06565. https://arxiv.org/abs/2212.06565
[15]	Gupta, A.; Singh, A. Agri-GNN: A Novel Genotypic-Topological Graph Neural Network Framework Built on GraphSAGE for Optimized Yield Prediction. arXiv:2310.13037. https://arxiv.org/abs/2310.13037
[16]	Bagheri, S.; Cheung, G.; Eadie, T. Graph Sparsification for GCN Towards Optimal Crop Yield Predictions. arXiv:2306.01725. https://arxiv.org/abs/2306.01725
[17]	Ma, M. et al. HiSTGNN: Hierarchical Spatio-temporal Graph Neural Networks for Weather Forecasting. arXiv:2201.09101. https://arxiv.org/abs/2201.09101
[18]	Chaudhary, M.; Gastli, M. S.; Nassar, L.; Karray, F. Deep Learning Approaches for Forecasting Strawberry Yields and Prices Using Satellite Images and Station-Based Soil Parameters. arXiv:2102.09024. https://arxiv.org/abs/2102.09024
[19]	Chang, S. et al. Learning Production Functions for Supply Chains with Graph Neural Networks. arXiv:2407.18772. https://arxiv.org/abs/2407.18772
[20]	West, R. Calibration of Google Trends Time Series. arXiv:2007.13861. https://arxiv.org/abs/2007.13861
[21]	Djorno, C.; Santillana, M.; Yang, S. Restoring the Forecasting Power of Google Trends with Statistical Preprocessing. arXiv:2504.07032. https://arxiv.org/abs/2504.07032
[22]	France, S. L.; Shi, Y. Aggregating Google Trends: Multivariate Testing and Analysis. arXiv:1712.03152. https://arxiv.org/abs/1712.03152
[23]	Challet, D.; Bel Hadj Ayed, A. Do Google Trend Data Contain More Predictability Than Price Returns? arXiv:1403.1715. https://arxiv.org/abs/1403.1715